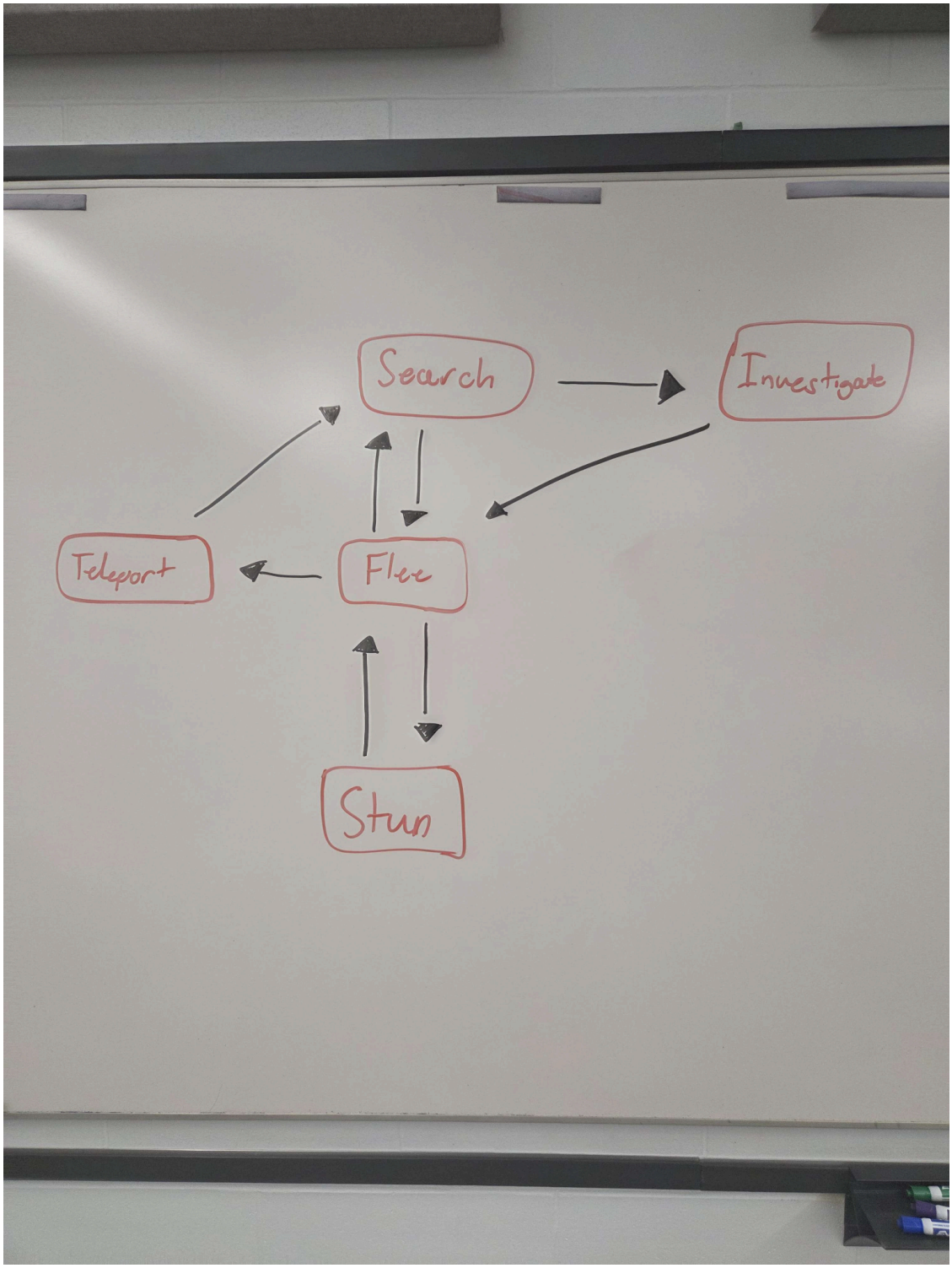
Youtube link to video: https://youtu.be/12ycQZT2L2U

AI State Machine Diagram



Search

Much like a patrol the Witch AI is given several search points to follow and rotate though.
They broadcast a visual cone when searching, it's narrow to allow the player to maneuver around them and adjust their direction when in the Flee state

- Search -> Investigate: Searchpoint reached
- Search -> Flee: Player in viewcone

Investigate

Almost like an Idle state, the Witch stops to check out the area around the searchPoint. Though unseen (damn gizmos), the Witch should have a radius around them where they're generally aware, this turns off in that state but the visual cone remains.

If they see the player, they'll flee and take those options.
If they don't they'll still go back to searching for their next waypoint.

- Investigate -> Search: Investigating time elapsed, onto the next searchpoint
- Investigate -> Flee: Player in viewcone

Flee

A state where the witch will take several actions to flee from the player.

If the player is on the outer half of the viewcone, the witch just turns tail and runs. They can teleport, but that will also go on a cooldown of several seconds once it happens once. But if the witch sees the players on the inner half of their view cone they teleport if able.

Otherwise they may stun the player.

Teleport and Stun can only be accessed through this state.

- Flee -> Search: Flee time elapsed, player not in viewcone
- Flee -> Teleport: Player in front half of viewcone, teleport not on cooldown
- Flee -> Stun: Player in front half of viewcone, teleport on cooldown

Teleport

A short lived state that can only be attained from the Flee state (kind of like a method of it) where the witch displaces herself and then goes into the search state (with assumed safety).

If the player is close enough when the witch teleports she may do it again.
The player has to be within the front/inner half of the viewcone for this action to take place.

- Teleport -> Search: New XZ position, player not in viewcone

Stun

The player's movement speed is set to 0 for a short time. This can only be done if the witch sees the player on the *inner half* of the viewcone and their teleport function is unavailable/on cooldown.

This would set the witch back to flee and finally to search.

Stun -> Flee: Teleport on cooldown, player in viewcone.

Challenges and Implementation Struggles

I think a big struggle I had during it was randomizing the searchpoint array which I ultimately had to abandon. As when Search was active, it kept randomizing them *constantly* so the AI wouldn't pick anywhere to go. So that was at least an hour and a half down the drain. Otherwise it was making sure that the AI wouldn't teleport outside of bounds or into terrain, which luckily just never seemed to happen.

Moreover I had to re-reference the patrol state from the example to make sure my searchpoints were correct in their logic, and the outcome was admittedly shoddy at best.

I did debate and dabble with disabling the player controller for the Stun state but that would lead to troubles with when and where to re-enable it and if it would even work, so I went with setting the player movespeed to 0 for a short time instead within the state and giving the witch reference to the player.

I did gradually lose count and sight of how many variables I had and what they did, so I did my best to organize them and even full on ctrl+f searched through my document to

make sure I used what I already had. Even now I’m not sure if I got everything organized and settled but I did comment on every major block I felt important.